

WED. Feb. 26-2025. Continuation of example of thrust chamber
(Section 4)

$$\bar{M}_{m_2})_{\text{nozzle exit}} = \sum X_i \bar{M}_i = 0.1947 \times 32 + 0.5369 \times 18 + 0.2684 \times 28 = \boxed{23.41}$$

$$\bar{R}_{m_2})_{\text{nozzle exit}} = \frac{\bar{R}}{\bar{M}_{m_2}} = \boxed{355.16 \frac{\text{J}}{\text{kgK}}}$$

$$\bar{c}_{p_{O_2}} = \bar{c}_{p_{N_2}} = \frac{\gamma_{d,i} \bar{R}}{\gamma_{d,i} - 1} = \boxed{4.4483 \bar{R}}$$

$$\bar{c}_{p_{H_2O}} = \frac{\gamma_{tr,i} \bar{R}}{\gamma_{tr,i} - 1} = \boxed{6.8824 \bar{R}}$$

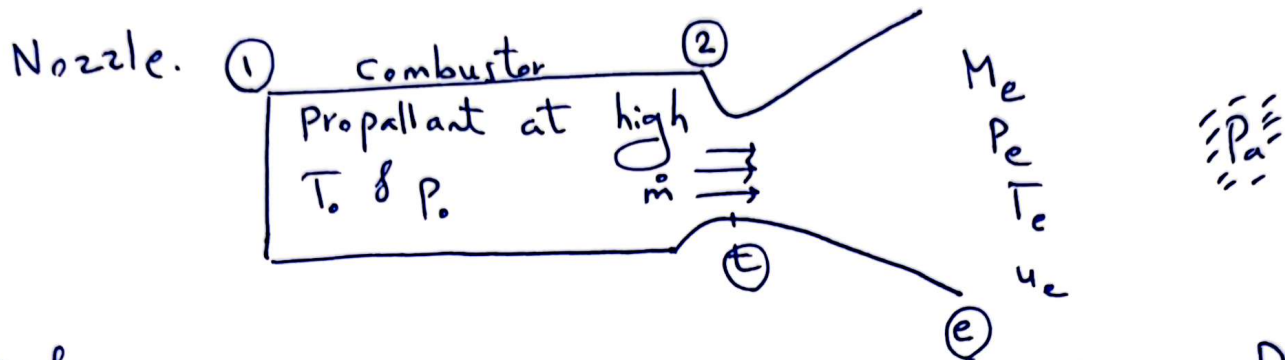
$$\begin{aligned} \bar{c}_{p_{m_2}} &= \sum X_i \bar{c}_{p_i} = \underbrace{0.1947 \times 4.4483 \bar{R}}_{X_{O_2} \times \bar{c}_{p_{O_2}}} + \underbrace{0.2684 \times 4.4483 \bar{R}}_{X_{N_2} \times \bar{c}_{p_{N_2}}} \\ &\quad + \underbrace{0.5369 \times (6.8824 \bar{R})}_{X_{H_2O} \times \bar{c}_{p_{H_2O}}} = \boxed{5.7552 \bar{R}} \end{aligned}$$

$$\gamma_{m_2} = \frac{\bar{c}_{p_{m_2}}}{\bar{c}_{p_{m_2}} - \bar{R}} = \boxed{1.21}$$

chemical Rocket Thrust chamber (Ch. II of text book)

(Section 6)

Thrust chamber consists of combustion chamber & Nozzle.



Define :

1) characteristic Velocity :

$$c^* = \frac{P_o A^*}{\dot{m}}$$

Dimension

$$\left[\frac{m}{s} \right]$$

2) Thrust coefficient :

$$c_z = \frac{\tau}{P_o A^*}$$

→ Thrust

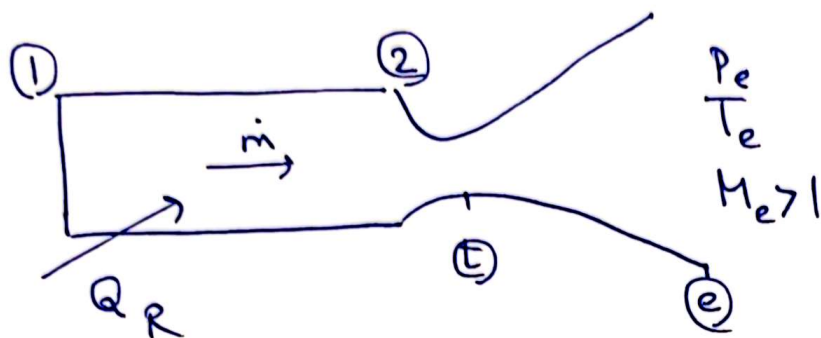
$$[-]$$

$$\Rightarrow \tau = c_z P_o A^* = c_z \dot{m} c^* \Rightarrow$$

$$\tau = \dot{m} c^* c_z$$

Assumptions of ideal thrust chamber

- ① propellant is perfect gas without ~~comp~~ constant composition.
- ② combustion is equivalent to ideal constant pressure ($p_o = cte, M = 0$) heat addition.
- ③ Nozzle is steady, 1D, isentropic.



Q_R : "heat added per unit mass of propellant"

1st law in combustion chamber: $c_p T_{o1} + Q_R = c_p T_{o2}$

$$\Rightarrow T_{o2} = T_{o1} + \frac{Q_R}{c_p} \quad (0)$$

Note that $T_{o2} = T_{oe}$

1st law in nozzle ($u_2 = 0$)

$$c_p T_{o2} = \frac{u_e^2}{2} + c_p T_e \quad (2)$$

$$\Rightarrow u_e = \sqrt{2c_p T_{o2} \left(1 - \frac{T_e}{T_{o2}}\right)} \quad (1)$$

Isentropic relation

$$\frac{T_e}{T_{o2}} = \left(\frac{P_e}{P_{o2}}\right)^{\frac{\gamma-1}{\gamma}} \quad (2)$$

(1) & (2) gives:

$$u_e = \sqrt{2c_p T_{o2} \left[1 - \left(\frac{P_e}{P_{o2}}\right)^{\frac{\gamma-1}{\gamma}}\right]} \quad (*)$$

$$u_e = \sqrt{\frac{2\gamma\bar{R}}{(\gamma-1)\bar{M}} T_{o2} \left[1 - \left(\frac{P_e}{P_{o2}}\right)^{\frac{\gamma-1}{\gamma}}\right]} \quad (*1)$$

From (0) & (*) $\rightarrow u_e = \sqrt{2c_p \left(\underbrace{T_{o1} + \frac{Q_R}{c_p}}_{T_{oe}}\right) \left[1 - \left(\frac{P_e}{P_{o2}}\right)^{\frac{\gamma-1}{\gamma}}\right]}$

(P-21)

$$T_{o_1} \ll \frac{Q_R}{c_p} \Rightarrow T_{o_1} + \frac{Q_R}{c_p} \approx \frac{Q_R}{c_p}$$

$$u_e \approx \sqrt{2 Q_R \left[1 - \left(\frac{P_e}{P_{o_2}} \right)^{\frac{\gamma-1}{\gamma}} \right]} = \sqrt{2 \frac{\bar{Q}_R}{\bar{M}} \left[1 - \left(\frac{P_e}{P_{o_2}} \right)^{\frac{\gamma-1}{\gamma}} \right]}$$

How to increase u_e ?

- ① Higher $\bar{Q}_R$
- ② lower mass of propellant $\bar{M}$
- ③ Higher P_{o_2}

All corresponds to combustion chamber

lowering P_e is also good but needs higher $\left(\frac{A_e}{A^*} \right)$

(nozzle change)

Note $\dot{m} = \frac{P_{o_2} A^*}{\sqrt{R T_{o_2}}} \sqrt{\gamma \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}}}$

(*)

$$\tau = \dot{m} u_e + (P_e - P_a) A_e$$

$$C_\tau = \frac{\tau}{P_o A^*} = \frac{\dot{m} u_e}{P_o A^*} + \left(\frac{P_e}{P_o} - \frac{P_a}{P_o} \right) \frac{A_e}{A^*}$$

replace u_e from (*) & $\dot{m}$ from (**)

$$C_\tau = \sqrt{\frac{2 \gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}}} \left[1 - \left(\frac{P_e}{P_o} \right)^{\frac{\gamma-1}{\gamma}} \right] + \left(\frac{P_e}{P_o} - \frac{P_a}{P_o} \right) \frac{A_e}{A^*}$$

(P-31)

Note: $\frac{A}{A^*} = f(M_e)$ & $\frac{P_e}{P_o} = f(M_e) = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-\frac{\gamma}{\gamma-1}}$

c^* implicitly depends on M_e ! How to find M_e ?

① If $\frac{A_e}{A^*}$ is given, find M_e from online calculator

② If pressure ratio at the nozzle exit is given, find

M_e , i.e.

$$\frac{P_e}{P_{o_e}} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-\frac{\gamma}{\gamma-1}}$$

(A)

Note $P_{o_e} = P_{o_2} = P_o$, $T_{o_e} = T_{o_2} = T_{o_1} + \frac{Q_R}{C_p}$

use $\left\{ \begin{array}{l} c^* = \frac{P_o A^*}{\dot{m}} \\ \dot{m} = \frac{P_o A^*}{\sqrt{R T_{o_2}}} \sqrt{\gamma \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}} \end{array} \right\}$ gives the following Eqⁿ for c^*

$$c^* = \sqrt{\frac{1}{\gamma} \left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{\gamma-1}} \frac{\bar{R} T_{o_2}}{\bar{M}}}$$

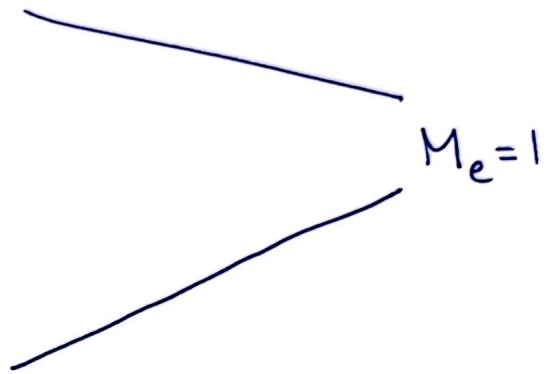
$c^* = c^*(\gamma, \bar{R}, \bar{M}, T_{o_2}) \rightarrow$ function of combustion chamber properties

Consider a convergent nozzle ($M_e = 1$)

From (A) for pressure ratio
with $M_e = 1$

$$\frac{P_e}{P_o} = \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1}}$$

(B)



If we put (B) in C_z Eq., we get $c_{z_{conv}}$

$$c_{z_{conv}} = (\gamma + 1) \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1}} - \frac{P_a}{P_o}$$